A Cheatsheet for World Models

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Chapter 1

Energy-Based Model

This chapter follows [1]. Consider a model with two sets of variables $\begin{cases} X & \text{the observed variables} \\ Y & \text{variables to be predicted} \end{cases}$. Variable X could be a vector containing the pixels from an image of an object. Variable Y could be a discrete variable that represents the possible category of the object. The model is viewed as an energy function which measures the goodness of each possible configuration of X and Y, denoted $E(Y,X)$. The output number can be interpreted as the degree of compatibility between the values of X and Y. Then the problem becomes

$$\min_{Y \in \mathcal{Y}} E(Y,X)$$

where the set $\mathcal{Y}$ needs to be defined. If the size of $\mathcal{Y}$ is too big, it can be impractical to do brute-force search, while if the size is too small, it may not contain the true Y.

The inference procedure means finding the Y that minimises $E(Y,X)$. The best inference procedure to use depends on the internal structure of the model. For example, if $\mathcal{Y}$ is continuous and $E(Y,X)$ is smooth and well-behaved w.r.t. Y, one may use a gradient-based optimisation algorithm. If Y is a collection of discrete variables and the energy function can be expressed as a factor graph, efficient inference procedures for factor graphs can be used.

For decision-making tasks, the system must give the lowest energy to the correct answer. The energies of other answers are irrelevant as long as they are larger. But since energies are uncalibrated, combining two separately trained energy-based models is not straightforward. Calibrating energies in order to permit such combinations can be done in multiple ways, but the only consistent way involves turning the collection of energies for all possible outputs into a normalised probability distribution. For example, through Gibbs distribution

$$P(Y|X) = \frac{e^{-\beta E(Y,X)}}{\int_{y \in \mathcal{Y}} e^{-\beta E(y,X)} dy}$$

where β is an arbitrary positive constant akin to an inverse temperature and the denominator is called the partition function.

1.1 Energy-Based Training

1.1.1 Loss Functions

Training an EBM consists in finding an energy function that produces the best Y for any X. The search for the best energy function is performed within a family of energy functions $\mathscr{E}$ indexed by a parameter W, $\mathscr{E} = \{E(W,Y,X) : W \in \mathscr{W}\}$.

Definition 1.1.1. *The architecture of the EBM is the internal structure of the parametrised energy function $E(W,Y,X)$.*

To train the model, we are given a set of training samples $\mathscr{S} = \{(X^i, Y^i)\}$ where X^i is the input for the ith training sample and Y^i is the corresponding desired answer. In order to find the best energy function in the family $\mathscr{E}$, we need to assess the quality of any particular energy function, based solely on the training set and our prior knowledge about the task. The quality measure is called the loss function, denoted $\mathscr{L}(E,\mathscr{S})$. For simplicity we often denote it $\mathscr{L}(W,\mathscr{S})$ and simply call it the loss function. Then the problem can be re-written as

$$\min_{W \in \mathscr{W}} \mathscr{L}(W,\mathscr{S})$$

In most cases, the loss functional is defined as

$$\mathscr{L}(E,\mathscr{S}) = \frac{1}{P}\sum_{i=1}^{P} L(Y^i, E(W,\mathscr{Y},X^i)) + R(W)$$

where the first term is the average taken over the training set of a per-sample loss function, which depends on Y^i and on the energies obtained by keeping the input sample fixed and varying the answer Y, and $R(W)$ is the regulariser embodying the prior knowledge about which energy functions are preferable to others.

Intuitively, the per-sample loss functional should be designed in such a way that it assigns a low loss to well-behaved energy functions, i.e. energy functions that give the lowest energy to the correct answer and higher energy to all other answers. Conversely, energy functions that do not assign the lowest energy to the correct answers would have a high loss. Training an EBM consists in shaping the energy function, so that for any given X the inference algorithm will produce the desired value for Y. Since the inference algorithm selects the

Y with the lowest energy, the learning procedure must shape the energy surface so that the desired value of Y has lower energy than all other values. Here denote $\begin{cases} Y^i & \text{the correct answer} \\ Y^{*i} & \text{the answer produced by the model} \\ \bar{Y}^i & \text{the most offending incorrect answer} \end{cases}$. With a properly designed loss function, the learning process should have the effect of pushing down on $E(W,Y^i,X^i)$ and pulling up on the incorrect energies, particularly on $E(W,\bar{Y}^i,X^i)$.

e.g.

- energy loss

$$L_{\text{energy}}(W,Y^i,X^i) = E(W,Y^i,X^i)$$

  With this loss function, the wrong answers are not pulled up on and therefore with some architectures, this can lead to a collapsed solution in which the energy is constant and equal to zero. Thus the energy loss will only work with architectures that are designed in such a way that pushing down on $E(W,Y^i,X^i)$ will automatically make the energies of the other answers higher, e.g. $E(W,Y^i,X^i) = ||Y^i - G(W,X^i)||^2$, where G is the regression function.

- generalised perceptron loss

$$L_{\text{perc}}(W,Y^i,X^i) = E(W,Y^i,X^i) - \min_{y \in \mathcal{Y}} E(W,y,X^i)$$

  which is always non-negative since the second term is a lower bound on the first term. Minimising this loss has the effect of pushing down on $E(W,Y^i,X^i)$ while pulling up on the energy of the answer produced by the model, Y^{*i}. Since it has no mechanism for creating an energy gap between the correct answer and the incorrect ones, it may produce flat energy surfaces if the architecture allows it.

- generalised margin loss

  Definition 1.1.2. *Let Y be a $\begin{cases} \text{discrete} \\ \text{continuous} \end{cases}$ variable. Then for a training sample (X^i,Y^i), the most offending incorrect answer $\bar{Y}^i$ is the answer that has the lowest energy among $\begin{cases} \text{all answers that are incorrect} \\ \text{all answers that are at least } \epsilon \text{ away from the correct answer} \end{cases}$;*

$$\bar{Y}^i = \arg\min_{Y \in \mathcal{Y}} E(W,Y,X^i), \quad \begin{cases} Y \neq Y^i \\ ||Y - Y^i|| > \epsilon \end{cases}$$

The generalised margin loss uses the energy of the most offending incorrect answer in the contrastive term

$$L_{\text{margin}}(W,Y^i,X^i) = Q_m(E(W,Y^i,X^i),E(W,\bar{Y}^i,X^i))$$

where m is a positive parameter called the margin and $Q_m(e_1,e_2)$ is a convex function whose gradient has a positive dot product with the vector $[1,-1]$ in the region where $E(W,Y^i,X^i)+m > E(W,\bar{Y}^i,X^i)$, i.e. the loss surface is slanted toward low values of $E(W,Y^i,X^i)$ and high values of $E(W,\bar{Y}^i,X^i)$ wherever $E(W,Y^i,X^i)$ is not smaller than $E(W,\bar{Y}^i,X^i)$ by at least m.

- negative log-likelihood loss

$$L_{\text{nll}}(W,Y^i,X^i) = E(W,Y^i,X^i) + \mathscr{F}_\beta(W,\mathscr{Y},X^i)$$

where $\mathscr{F}$ is the free energy of the ensemble $\{E(W,y,X^i),\ y\in\mathscr{Y}\}$ s.t.

$$\mathscr{F}_\beta(W,\mathscr{Y},X^i) = \frac{1}{\beta}\log(\int_{y\in\mathscr{Y}} \exp(-\beta E(W,y,X^i))dy)$$

where β is a positive constant akin to an inverse temperature.
The form of the nll loss function stems from a probabilistic formulation of the learning problem in terms of the maximum conditional probability principle where observations are independent.

$$\begin{array}{ll} P(Y^1,\ldots,Y^P|X^1,\ldots,X^P,W) = \prod_{i=1}^P P(Y^i|X^i,W) & \\ \leftrightarrow -\log\prod_{i=1}^P P(Y^i|X^i,W) = \sum_{i=1}^P -\log P(Y^i|X^i,W) & \max f \text{ is equivalent to } \min -f \\ = \sum_{i=1}^P (\beta E(W,Y^i,X^i) + \log\int_{y\in\mathscr{Y}} e^{-\beta E(W,y,X^i)}dy) & \text{by the Gibbs distribution} \end{array}.$$

Therefore the loss function becomes

$$\mathscr{L}_{\text{nll}}(W,\mathscr{S}) = \frac{1}{P}\sum_{i=1}^P (E(W,Y^i,X^i) + \frac{1}{\beta}\log\int_{y\in\mathscr{Y}} e^{-\beta E(W,y,X^i)}dy)$$

$\rightarrow \frac{\partial L_{\text{nll}}(W,Y^i,X^i)}{\partial W} = \frac{\partial E(W,Y^i,X^i)}{\partial W} - \int_{y\in\mathscr{Y}} \frac{\partial E(W,y,X^i)}{\partial W} P(y|X^i,W)dy$ where $P(Y|X^i,W) = \frac{e^{-\beta E(W,Y,X^i)}}{\int_y e^{-\beta E(W,y,X^i)}dy}$ i.e. the contrastive term causes the energies of all the answers to be pulled up, even the correct answer, in proportion to the likelihood of that answer under the model.

When $\beta\rightarrow\infty$, the NLL loss reduces to the generalised perceptron loss.

1.1.2 Architecture

Consider the case where the energy functions depend on a set of hidden variables Z whose correct value is never given to us, even during training. For these

architectures, the inference process for a given set of variables X, Y involves minimising over these unseen variables Z

$$E(Y,X) = \min_{Z \in \mathcal{Z}} E(Z,Y,X)$$

Such hidden variables are called latent variables. Even though the evaluation of $E(Y,X)$ involving a minimisation over Z does not significantly impact the approach described so far, the special treatment is necessary, because of a subtlety: unlike Y, Z is never observed (see below). Consider the simultaneous minimisation over Y and $Z \rightarrow Y^* = \arg\min_{Y \in \mathcal{Y}, Z \in \mathcal{Z}} E(Z,Y,X)$. Latent variables can be viewed as intermediate results on the way to finding the best output Y. Z and Y may seem indistinguishable, since Z could be folded into Y. But during training we are given the correct value of Y for training samples while we are never given the correct value of Z.

e.g.

- face detection

  Consider the case with a face detecting function $G_{face}(X)$ which takes a small image window as input and produces a scalar output. This output is a small value when a human face fills the input image, and a large value if no face is present. The variable Y controls the position of a binary switch $\begin{cases} \text{face} & 1 \\ \text{non-face} & 0 \end{cases}$. The output energy is equal to $\begin{cases} G_{face}(X) \text{ when } Y = 1 \\ T \text{ when } Y = 0 \end{cases} \rightarrow E(Y,X) = YG_{face}(X) + (1-Y)T$ where T denotes the threshold. The value of Y that minimises the energy function is $\begin{cases} 1 & G_{face}(X) < T \\ 0 & \text{otherwise} \end{cases}$.

- face detection and location

  Here we can apply $G_{face}(X)$ function to multiple windows in the large image, compute which window produces the lowest value of $G_{face}(X)$, and detect a face at that location if the value is lower than T. The latent location variable Z selects which of the K copies of the G_{face} function is routed to the output energy. Then the energy function becomes $E(Z,Y,X) = Y(\sum_{k=1}^{K} \delta(Z-k) G_{face}(X_k)) + (1-Y)T$ where the X_ks are the image windows. Locating the best-scoring location in the image consists in minimising the energy w.r.t. Y, Z. The resulting value of Y will indicate whether a face was found and the resulting value of Z will indicate the location.

When latent variables are present, the joint conditional distribution over Y, Z given by the Gibbs distribution is

$$P(Z,Y|X) = \frac{e^{-\beta E(Z,Y,X)}}{\int_{y,z} e^{-\beta E(z,y,X)} dz\, dy}$$

$\rightarrow P(Y|X) = \frac{\int_z e^{-\beta E(z,Y,X)} dz}{\int_{y,z} e^{-\beta E(z,y,X)} dz\, dy}$ by marginalising over Z. Then given the denominator is a constant in Y, finding the best Y reduces to

$$Y^* = \arg\min_{Y \in \mathcal{Y}} -\frac{1}{\beta} \log \int_z e^{-\beta E(z,Y,X)} dz$$

If $\beta \rightarrow \infty$ (zero temperature), the inference formula reduces to the previous simultaneous optimisation formula.

Remark. IV/2SLS uses instruments which are observable so the estimation occurs in stages, while here the latent variables are unobservable and the optimisation occurs simultaneously.

1.1.3 Good Loss Functions for EBM

Condition 1. For sample (X^i, Y^i), the machine will give the correct answer for X^i if

$$E(W,Y^i,X^i) < E(W,Y,X^i),\ \forall Y \in \mathcal{Y},\ Y \neq Y^i$$

$\rightarrow$ the condition for the energy function to be correct.

Condition 2. For a variable Y and sample (X^i, Y^i) and positive margin m, the inference algorithm will give the correct answer for X^i if

$$E(W,Y^i,X^i) < E(W,\bar{Y}^i,X^i) - m$$

$\rightarrow$ the condition for the energy function to be correct and robustly stable with margin m.

If the system is to produce the correct answers, the loss functional should be designed in such a way that minimising it will cause $E(W,Y^i,X^i)$ to be lower than $E(W,\bar{Y}^i,X^i)$ by some margin m. Since only the relative values of those two energies matter, we only need to consider the shape of a slice of the loss functional in the 2D space of those two energies. For example, in the case where $\mathcal{Y}$ is

the set of integers from 1 to k, the loss functional can be written as $L(W,Y^i,X^i) = L(Y^i,E(W,1,X^i),\ldots,E(W,k,X^i))$. The projection of this loss in the space of $E(W,Y^i,X^i)$ and $E(W,\bar{Y}^i,X^i)$ can be viewed as a function Q parametrised by the other $k-2$ energies

$$L(W,Y^i,X^i) = Q_{[E_y]}(E(W,Y^i,X^i),E(W,\bar{Y}^i,X^i))$$

where $[E_y]$ contains the vector of energies for all values of Y except Y^i and $\bar{Y}^i$. Assume the existence of at least one set of parameters W for which condition 2 is satisfied for a single training sample (X^i,Y^i). If no such W exists, there is no point in considering the model since there cannot exist any loss function satisfying condition 2. For simplicity denote the energy $\begin{cases} E(W,Y^i,X^i) & E_C \\ E(W,\bar{Y}^i,X^i) & E_I \end{cases}$. Then we can get a 3-dimensional plot of the square-square loss function in which the abscissa is E_C and the ordinate is E_I. The third axis gives the value of the loss for the corresponding values of E_C,E_I. Then the loss function is a family of 2D surfaces in this 3D space, where each surface corresponds to one particular configuration of all energies except E_C and E_I.

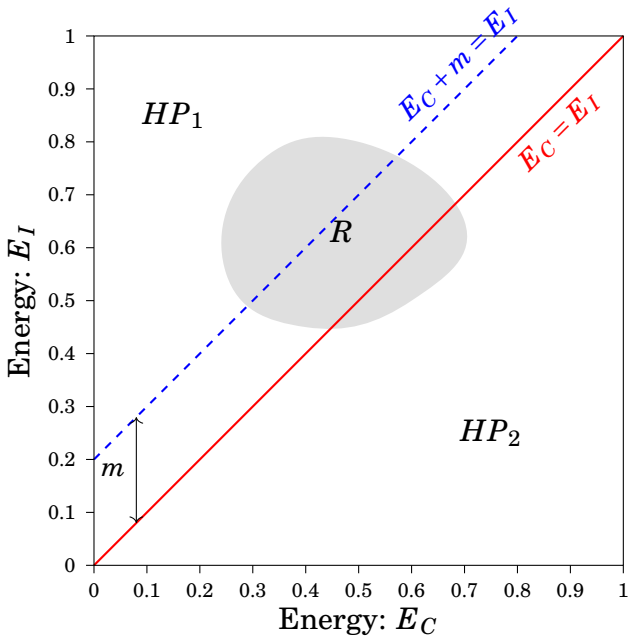


Figure 1.1: The various regions in the plane of the two energies E_C and E_I. E_C are the (correct answer) energies associated with (X^i,Y^i), and E_I are the (incorrect answer) energies associated with $(X^i,\bar{Y}^i)$ [1].

The solid red line in the figure corresponds to the points in the 2D plane for which $E_C = E_I$. The dashed blue line corresponds to the margin line $E_C + m = E_I$.

Let the two half planes $E_C + m < E_I$ and $E_C + m \geq E_I$ be denoted by HP_1 and HP_2 respectively. Let R be the feasible region, defined as the set of values (E_C, E_I) corresponding to all possible values of $W \in \mathcal{W}$. This region may be non-convex, discontinuous, open or one-dimensional and could lie anywhere in the plane. In the figure R is represented as the shaded region. Then by the assumption, a solution exists which satisfies the condition 2, and therefore R must intersect HP_1.

Let two points $\begin{cases} (e_1, e_2) \\ (e'_1, e'_2) \end{cases}$ belong to the feasible region R, s.t. $\begin{cases} (e_1, e_2) \in HP_1 \\ (e'_1, e'_2) \in HP_2 \end{cases}$.

Condition 3 (sufficient condition). Let (X^i, Y^i) be the ith training example and m be a positive margin. Minimising the loss function L will satisfy conditions 1 or 2 if there exists at least one point (e_1, e_2) with $e_1 + m < e_2$ s.t. $\forall (e'_1, e'_2)$ with $e'_1 + m \geq e'_2$,

$$Q_{[E_y]}(e_1, e_2) < Q_{[E_y]}(e'_1, e'_2) \textbf{ where } L(W, Y^i, X^i) = Q_{[E_y]}(E(W, Y^i, X^i), E(W, \bar{Y}^i, X^i))$$

$\rightarrow$ the surface of the loss function in the space of E_C, E_I should be s.t. there exists at least one point in the part of the feasible region R intersecting HP_1 s.t. the value of the loss function at this point is less than its value at all other points in the part of R intersecting HP_2.

e.g.

- negative log-likelihood loss

  For a fixed parameter W and a sample (X^i, Y^i), consider the gradient of the loss w.r.t. the energy E_C of the correct answer Y^i and the energy E_I of the most offending incorrect answer $\bar{Y}^i$. Then we have

$$\begin{cases} g_C = \frac{\partial L(W, Y^i, X^i)}{\partial E_C} = 1 - \frac{e^{-E(W, Y^i, X^i)}}{\sum_Y e^{-E(W, Y, X^i)}} \\ g_I = \frac{\partial L(W, Y^i, X^i)}{\partial E_I} = -\frac{e^{-E(W, \bar{Y}^i, X^i)}}{\sum_Y e^{-E(W, Y, X^i)}} \end{cases}$$

  Then for any value of energies, $\begin{matrix} g_C > 0 \\ g_I < 0 \end{matrix}$.

  Claim. There exists at least one point in HP_1 at which the loss is less than at all the points in HP_2.

  pf. Let $A = (E^*_C, E^*_C + m)$ be a point on the margin line for which the loss is minimum. E^*_C is the value of the correct energy at this point, i.e. $E^*_C = \arg\min Q_{[E_y]}(E_C, E_C + m)$. The negative of the gradient of the loss $Q_{[E_y]}$ points, at all points, in a direction inside HP_1 by the argument about g above. Then by the monotonicity of the loss, $Q_{[E_y]}(E^*_C, E^*_C + m) \leq Q_{[E_y]}(E_C, E_I)$

where $E_C + m \geq E_I$. Consider a point B at a distance ϵ away from the point $(E_C^*, E_C^* + m)$ and inside HP_1. That is the point $(E_C^* - \frac{\epsilon}{\sqrt{2}}, E_C^* + m + \frac{\epsilon}{\sqrt{2}})$. By the first order Taylor expansion at this point, we get

$$\begin{aligned} Q_{[E_y]}(E_C^* - \tfrac{\epsilon}{\sqrt{2}}, E_C^* + m + \tfrac{\epsilon}{\sqrt{2}}) &= Q_{[E_y]}(E_C^*, E_C^* + m) - \tfrac{\epsilon}{\sqrt{2}} \frac{\partial Q_{[E_y]}}{\partial E_C} + \tfrac{\epsilon}{\sqrt{2}} \frac{\partial Q_{[E_y]}}{\partial E_I} + O(\epsilon^2) \\ &= Q_{[E_y]}(E_C^*, E_C^* + m) + \tfrac{\epsilon}{\sqrt{2}} [\frac{\partial Q_{[E_y]}}{\partial E_C}, \frac{\partial Q_{[E_y]}}{\partial E_I}] \begin{bmatrix} -1 \\ 1 \end{bmatrix} + O(\epsilon^2) \end{aligned}$$

where the second term on the RHS is negative. So for sufficiently small ϵ, $Q_{[E_y]}(E_C^* - \frac{\epsilon}{\sqrt{2}}, E_C^* + m + \frac{\epsilon}{\sqrt{2}}) < Q_{[E_y]}(E_C^*, E_C^* + m)$ and therefore there exists at least one point in HP_1 at which the loss is less than all points in HP_2.

1.2 Efficient Inference

Chapter 2

VICReg

Chapter 3

The World Model

Appendix A

Graph Laplacian and Spectral Clustering

This appendix follows [2].

A.1 Preliminaries

Given a set of data points $x_1,\ldots,x_n$ and some notion of similarity $s_{ij} \geq 0$ between all pairs of data points x_i, x_j, the weighted adjacency matrix is $W = (w_{ij})$ where $w_{ij} = \begin{cases} s_{ij} & s_{ij} > t \\ 0 & \text{otherwise} \end{cases}$. Form the similarity graph $G = (V,E)$, where $\begin{cases} v_i & \text{a data point } x_i \\ e_{ij} & \begin{cases} 1 & s_{ij} > t \\ 0 & \text{otherwise} \end{cases} \end{cases}$ where t is a certain threshold. Then the problem of clustering can be reformulated to the problem of partitioning graph s.t. the edges between different groups have very low weights and the edges within a group have high weights. Assume G is undirected, i.e. $W = W^T$, $\forall i,j,\ w_{ij} = w_{ji}$. The degree of a vertex $v_i \in V$ is defined as $d_i = \sum_{j=1}^{n} w_{ij}$. The degree matrix D is defined as the diagonal matrix with degrees $d_1,\ldots,d_n$ on the diagonal. Given a subset of vertices $A \subset V$, $\bar{A} \equiv V \setminus A$. Define the indicator vector $\mathbb{1}_A = (f_1,\ldots,f_n)^T \in \mathbb{R}^n$ as the vector with $f_i = \begin{cases} 1 & v_i \in A \\ 0 & v_i \in \bar{A} \end{cases}$. Also define the size of a subset $A \subset V$, as the number of vertices in A, denoted $|A|$ and $\text{vol}(A) \equiv \sum_{i \in A} d_i$. A subset A is called connected if any two vertices in A can be joined by a path s.t. all intermediate vertices also lie in A. A subset A is called a connected component if it is connected and if there are no connections between vertices in A and $\bar{A}$. For two not necessarily disjoint sets $A, B \subset V$, $W(A,B) \equiv \sum_{i \in A, j \in B} w_{ij}$. The nonempty sets $A_1,\ldots,A_k$ form a partition of the graph if $\begin{cases} A_i \cap A_j = \emptyset \\ A_1 \cup \cdots \cup A_k = V \end{cases}$.

There are several ways to transform a given set $x_1, \ldots, x_n$ of data points with pairwise similarities s_{ij} into a graph. The most widely used ones include the ϵ-neighbourhood graph, k-nearest neighbour graph, and the fully connected graph.

A.2 Un-normalised Case

The unnormalised graph Laplacian matrix is defined as $L = D - W$.

Proposition A.2.1. *The matrix L satisfies the following properties:*

1. $\forall f \in \mathbb{R}^n,\ f^T L f = \frac{1}{2}\sum_{i,j=1}^n w_{ij}(f_i - f_j)^2$
2. *L is symmetric and positive semi definite (p.s.d.).*
3. *The smallest eigenvalue of L is 0, the corresponding eigenvector is the constant one vector* $\mathbb{1}$.
4. *L has n non-negative real-valued eigenvalues* $0 = \lambda_1 \leq \lambda_2 \leq \cdots \leq \lambda_n$

pf.

- $$\begin{aligned} f^T L f &= f^T D f - f^T W f && \text{by the definition of } L \\ &= \sum_{i=1}^n d_i f_i^2 - \sum_{i,j=1}^n f_i f_j w_{ij} && \text{by the definition of } d_i \\ &= \frac{1}{2}(\sum_{i=1}^n d_i f_i^2 - 2\sum_{i,j} f_i f_j w_{ij} + \sum_{j=1}^n d_j f_j^2) \\ &= \frac{1}{2}\sum_{i,j} w_{ij}(f_i - f_j)^2 \end{aligned}$$.

- The symmetry of L follows directly from the symmetry of W, D. The p.s.d. is direct consequence of the first stmt, which shows $f^T L f \geq 0,\ \forall f \in \mathbb{R}^n$, given $w_{ij} \geq 0$.

- $L\mathbb{1} = D\mathbb{1} - W\mathbb{1} = (d_i - \sum_{j=1}^n w_{ij})_{i=1,\ldots,n} = 0$ by the definition of d_i, i.e. $\mathbb{1}$ is an eigenvector of L with eigenvalue 0. Since L is p.s.d., all its eigenvalues are non-negative. Thus the smallest eigenvalue of L is 0, with eigenvector $\mathbb{1}$.

- direct consequences of the previous stmts.

Proposition A.2.2. *Let G be an undirected graph with non-negative weights. Then the multiplicity k of the eigenvalue 0 of L equals the number of connected components* $A_1, \ldots, A_k$ *in the graph. The eigenspace of eigenvalue 0 is spanned by the indicator vectors* $\mathbb{1}_{A_1}, \ldots, \mathbb{1}_{A_k}$ *of those components.*

pf.

- $k = 1$

  the graph is connected. Assume f is an eigenvector with eigenvalue 0. Then $0 = f^T L f = \sum_{ij} w_{ij}(f_i - f_j)^2$ where $w_{ij} \geq 0$, $\forall i, j$. Then it is only possible when $w_{ij}(f_i - f_j)^2$ vanish. Thus if two vertices v_i, v_j are adjacent, $f_i = f_j$. With this argument, f needs to be constant for all vertices which can be connected by a path in the graph. Additionally, as all vertices of a connected component in an undirected graph can be connected by a path, f needs to be constant on the whole connected component. Since we have only 1 component, we only have the constant one vector $\mathbb{1}$ as eigenvector with eigenvalue 0.

- $k \geq 2$

  Wlog, assume that the vertices are ordered according to the connected components they belong to. Then the adjacency matrix W has a block diagonal form and the same is true for the matrix L. Then each of the blocks L_i is a proper graph Laplacian on its own, namely the Laplacian corresponding to the subgraph of the ith connected component. As it is the case for all block diagonal matrices, the spectrum of L is given by the union of the spectra of L_i and the corresponding eigenvectors of L are the eigenvectors of L_i, filled with 0 at the positions of the other blocks. As each L_i is a graph Laplacian of a connected graph, we know each L_i has eigenvalue 0 with multiplicity 1, and the corresponding eigenvector is the constant one vector on the ith connected component. Thus the matrix L has as many eigenvalues 0 as there are connected components and the corresponding eigenvectors are the indicator vectors of the connected components.

We measure the pairwise similarities $s_{ij} = s(x_i, x_j)$ by some similarity function which is symmetric and non-negative and denote the corresponding similarity matrix by $S = (s_{ij})$. → The key trick used here is changing the representation of the abstract data points x_i to points $y_i \in \mathbb{R}^k$. This is possible due to the properties of the graph Laplacian.

A.3 Spectral Clustering as Graph Cut

The intuition of clustering is to separate points in different groups according to their similarities. For data given in the form of a similarity graph, it can be restated as to find a partition of the graph s.t. the edges between different groups have a very low weight and the edges within a group have high weight.

Algorithm 1 Unnormalised Spectral Clustering

Require: Similarity matrix $S \in \mathbb{R}^{n \times n}$, the number k of clusters to construct.
Ensure: Clusters $A_1, \ldots, A_k$ with $A_i = \{j \mid y_j \in C_i\}$

1: Construct a similarity graph by one of the ways described in Preliminaries. Let W be its weighted adjacency matrix.
2: Compute the unnormalised Laplacian L.
3: Compute the first k eigenvectors $u_1, \ldots, u_k$ of L.
4: Let $U \in \mathbb{R}^{n \times k}$ be the matrix containing the vectors $u_1, \ldots, u_k$ as columns.
5: **for** $i = 1, \ldots, n$ **do**
6: Let $y_i \in \mathbb{R}^k$ be the vector corresponding to the ith row of U.
7: **end for**
8: Cluster the points $(y_i)_{i=1,\ldots,n}$ in $\mathbb{R}^k$ with the k-means algorithm into clusters $C_1, \ldots, C_k$.

e.g. Mincut Problem
For a given number k of subsets, the mincut approach consists in choosing a partition $A_1, \ldots, A_k$ which minimises $\text{cut}(A_1, \ldots, A_k) \equiv \frac{1}{2} \sum_{i=1}^{k} W(A_i, \bar{A}_i)$.
→ when $k = 2$, it can be solved efficiently, but for $k > 2$, in many cases, the solution of mincut simply separates one individual vertex from the rest of the graph.

One solution for the above problem is forcing the sets $A_1, \ldots, A_k$ reasonably large.

- RatioCut

  The size of a subset A of a graph is measured by $|A|$ and the objective function becomes $\text{RatioCut}(A_1, \ldots, A_k) \equiv \frac{1}{2} \sum_{i=1}^{k} \frac{W(A_i, \bar{A}_i)}{|A_i|} = \sum \frac{\text{cut}(A_i, \bar{A}_i)}{|A_i|}$ where $|A_1| = \cdots = |A_k|$ the quantity's minimum is attained.

  - $k = 2$

    $$\min_{A \subset V} \text{RatioCut}(A, \bar{A}), \text{where } A \text{ is a proper subset of } V$$

    Given a nonempty subset $A \subset V$, define the vector $f = (f_1, \ldots, f_n)^T \in \mathbb{R}^n$ where $f_i = \begin{cases} \sqrt{|\bar{A}|/|A|} & v_i \in A \\ -\sqrt{|A|/|\bar{A}|} & v_i \in \bar{A} \end{cases}$. Then the objective function can be rewritten using the unnormalised graph Laplacian

$$
\begin{aligned}
f^T L f &= \tfrac{1}{2}\textstyle\sum_{ij} w_{ij}(f_i - f_j)^2 \\
&= \tfrac{1}{2}\textstyle\sum_{i\in A, j\in \bar{A}} w_{ij}\left(\sqrt{\tfrac{|\bar{A}|}{|A|}} + \sqrt{\tfrac{|A|}{|\bar{A}|}}\right)^2 + \tfrac{1}{2}\textstyle\sum_{i\in \bar{A}, j\in A} w_{ij}\left(-\sqrt{\tfrac{|\bar{A}|}{|A|}} - \sqrt{\tfrac{|A|}{|\bar{A}|}}\right)^2 && \text{by the definition of } f \\
&= \mathrm{cut}(A,\bar{A})\left(\tfrac{|\bar{A}|}{|A|} + \tfrac{|A|}{|\bar{A}|} + 2\right) && \text{by the definition of cut } W(A,\bar{A}) = W(\bar{A},A) \\
&= \mathrm{cut}(A,\bar{A})\left(\tfrac{|A|+|\bar{A}|}{|A|} + \tfrac{|A|+|\bar{A}|}{|\bar{A}|}\right) = |V|\mathrm{RatioCut}(A,\bar{A})
\end{aligned}
$$

Additionally $\sum_i f_i = \sum_{i\in A}\sqrt{\frac{|\bar{A}|}{|A|}} - \sum_{i\in\bar{A}}\sqrt{\frac{|A|}{|\bar{A}|}} = |A|\sqrt{\frac{|\bar{A}|}{|A|}} - |\bar{A}|\sqrt{\frac{|A|}{|\bar{A}|}} = 0$, i.e. the vector f is orthogonal to the constant one vector. Also $||f||^2 = \sum_i f_i^2 = |A|\frac{|\bar{A}|}{|A|} + |\bar{A}|\frac{|A|}{|\bar{A}|} = |\bar{A}| + |A| = n$. Thus the problem can be rewritten as

$$
\min_{A\subset V} f^T L f \text{ s.t. } \begin{cases} f \perp \mathbb{1} \\ ||f|| = \sqrt{n} \end{cases}, \text{ where } f_i = \begin{cases} \sqrt{|\bar{A}|/|A|} & v_i \in A \\ -\sqrt{|A|/|\bar{A}|} & v_i \in \bar{A} \end{cases}
$$

This is a discrete optimisation problem as the entries of the solution vector f are only allowed to take two particular values and it is NP-hard. So by discarding the discreteness condition, we can get the relaxed optimisation problem

$$
\min_{f\in\mathbb{R}^n} f^T L f \text{ s.t. } \begin{cases} f \perp \mathbb{1} \\ ||f|| = \sqrt{n} \end{cases}
$$

Then by the Rayleigh–Ritz theorem, the solution of this problem is given by the vector f which is the eigenvector corresponding to the second smallest eigenvalue of L, given that the smallest eigenvalue of L is 0 with eigenvector $\mathbb{1}$. Then we re-transform the real-valued solution vector f of the relaxed problem into a discrete indicator vector. The simplest way is to use the sign of f as indicator function, i.e. $\begin{cases} v_i \in A & f_i \geq 0 \\ v_i \in \bar{A} & f_i < 0 \end{cases}$. But this heuristic is too simple, and most spectral clustering algorithms use $\begin{cases} v_i \in A & f_i \in C \\ v_i \in \bar{A} & f_i \in \bar{C} \end{cases}$ where $C, \bar{C}$ are groups clustering f_i in $\mathbb{R}$.

– arbitrary k

Given a partition of V into k sets $A_1, \ldots, A_k$, we define k indicator vectors $h_j = (h_{1j}, \ldots, h_{nj})^T$ where $h_{ij} = \begin{cases} \frac{1}{\sqrt{|A_j|}} & v_i \in A_j \\ 0 & \text{otherwise} \end{cases}$. Then define the matrix $H \in \mathbb{R}^{n\times k}$ as the matrix containing those k indicator vectors as columns. Then the columns of H are orthonormal, i.e. $H^T H = I$.

Also $h_i^T L h_i = \frac{\text{cut}(A_i, \bar{A}_i)}{|A_i|} = (H^T L H)_{ii}$. This implies $\text{RatioCut}(A_1, \ldots, A_k) = \sum_{i=1}^k h_i^T L h_i = \sum (H^T L H)_{ii} = \text{Tr}(H^T L H)$. So the problem becomes

$$\min_{A_1,\ldots,A_k} \text{Tr}(H^T L H) \text{ s.t. } H^T H = I, \text{ where } h_{ij} = \begin{cases} \frac{1}{\sqrt{|A_j|}} & v_i \in A_j \\ 0 & \text{otherwise} \end{cases}$$

Then we can relax this problem by allowing the entries of the matrix H to take arbitrary real values, and get

$$\min_{H \in \mathbb{R}^{n \times k}} \text{Tr}(H^T L H) \text{ s.t. } H^T H = I$$

By the Rayleigh–Ritz theorem, the solution is given by choosing H as the matrix which contains the first k eigenvectors of L as columns. Here the matrix H is in fact the matrix U used in the unnormalised spectral clustering algorithm. Then we can re-convert the real-valued solution matrix to a discrete partition as previously.

A.4 Normalised Case

There are two matrices called normalised graph Laplacians:

$$\begin{cases} L_{sym} \equiv D^{-1/2} L D^{-1/2} = I - D^{-1/2} W D^{-1/2} \\ L_{rw} \equiv D^{-1} L = I - D^{-1} W \end{cases}.$$

Proposition A.4.1. *Given there is no isolated vertex, the normalised Laplacians satisfy the following properties:*

1. $\forall f \in \mathbb{R}^n,\ f^T L_{sym} f = \frac{1}{2} \sum_{i,j=1}^n w_{ij} (\frac{f_i}{\sqrt{d_i}} - \frac{f_j}{\sqrt{d_j}})^2$

2. *λ is an eigenvalue of L_{rw} with eigenvector u iff λ is an eigenvalue of L_{sym} with eigenvector $w = D^{1/2} u$.*

3. *λ is an eigenvalue of L_{rw} with eigenvector u iff λ and u solve the generalised eigenproblem $Lu = \lambda D u$.*

4. *0 is an eigenvalue of L_{rw} with the constant one vector $\mathbb{1}$ as eigenvector. 0 is an eigenvalue of L_{sym} with eigenvector $D^{1/2}\mathbb{1}$.*

5. *L_{sym} is p.s.d. and L_{sym},L_{rw} have n non-negative real-valued eigenvalues $0 = \lambda_1 \leq \cdots \leq \lambda_n$.*

pf.

- $$\begin{aligned} f^T L_{sym} f &= f^T f - f^T D^{-1/2} W D^{-1/2} f \quad \text{by the definition of } L_{sym} \\ &= \sum_{i=1}^n f_i^2 - \sum_{ij} \frac{f_i}{\sqrt{d_i}} \frac{f_j}{\sqrt{d_j}} w_{ij} \\ &= \frac{1}{2}(\sum f_i^2 - 2\sum_{ij} \frac{f_i f_j}{\sqrt{d_i d_j}} w_{ij} + \sum_j f_j^2) \\ &= \frac{1}{2} \sum_{ij} (\frac{f_j}{\sqrt{d_j}} - \frac{f_i}{\sqrt{d_i}})^2 w_{ij} \qquad \text{by the definition of } d_j \end{aligned}$$

- Multiplying the eigenvalue equation $L_{sym} w = \lambda w$ by $D^{-1/2}$ gives $D^{-1} L D^{-1/2} w = \lambda D^{-1/2} w$ which is equivalent to $D^{-1} L u = \lambda u$ where $u = D^{-1/2} w$.
- Multiplying the eigenvalue equation $L_{rw} u = \lambda u$ by D gives $Lu = \lambda D u$.
- $L_{rw} \mathbb{1} = 0$ gives the first stmt. The second stmt follows from stmt 2.
- Since L_{sym} is p.s.d. by stmt 1, its eigenvalues are all non-negative and by an argument similar to the unnormalised case, the stmt about L_{sym} holds. Then by the second stmt, the stmt holds for L_{rw}.
  $\rightarrow L_{rw}$ is not symmetric!

Proposition A.4.2. *Let G be an undirected graph with non-negative weights and with no isolated vertex. Then the multiplicity k of the eigenvalue 0 of both L_{rw} and L_{sym} equals the number of connected components $A_1, \ldots, A_k$ in the graph. For L_{rw}, the eigenspace of 0 is spanned by the indicator vectors $\mathbb{1}_{A_i}$ of those components. For L_{sym}, the eigenspace of 0 is spanned by the vectors $D^{1/2} \mathbb{1}_{A_i}$.*

pf. Similar to the unnormalised case.

Algorithm 2 Normalised Spectral Clustering

Require: Similarity matrix $S \in \mathbb{R}^{n \times n}$, the number k of clusters to construct.
Ensure: Clusters $A_1, \ldots, A_k$ with $A_i = \{j \mid y_j \in C_i\}$

1: Construct a similarity graph by one of the ways described in Preliminaries. Let W be its weighted adjacency matrix.
2: Compute the unnormalised Laplacian L.
3: Compute the first k generalised eigenvectors $u_1, \ldots, u_k$ of the generalised eigenproblem $Lu = \lambda D u$.
4: Let $U \in \mathbb{R}^{n \times k}$ be the matrix containing the vectors $u_1, \ldots, u_k$ as columns.
5: **for** $i = 1, \ldots, n$ **do**
6: Let $y_i \in \mathbb{R}^k$ be the vector corresponding to the ith row of U.
7: **end for**
8: Cluster the points $(y_i)_{i=1,\ldots,n}$ in $\mathbb{R}^k$ with the k-means algorithm into clusters $C_1, \ldots, C_k$.

$\rightarrow$ we can replace $Lu = \lambda D u$ with $L_{sym} w = \lambda w$ and get another version of normalised spectral clustering algorithm.

- NCUT

  The size of a subset A is measured by the $\mathrm{vol}(A)$ and the objective function becomes $\mathrm{NCut}(A_1,\dots,A_k) \equiv \frac{1}{2}\sum_{i=1}^{k} \frac{W(A_i,\bar{A}_i)}{\mathrm{vol}(A_i)} = \sum \frac{\mathrm{cut}(A_i,\bar{A}_i)}{\mathrm{vol}(A_i)}$ where $\mathrm{vol}(A_1) = \cdots = \mathrm{vol}(A_k)$ the quantity's minimum is attained.

  - $k = 2$

    Define the cluster indicator vector f where $f_i = \begin{cases} \sqrt{\frac{\mathrm{vol}(\bar{A})}{\mathrm{vol}(A)}} & v_i \in A \\ -\sqrt{\frac{\mathrm{vol}(A)}{\mathrm{vol}(\bar{A})}} & v_i \in \bar{A} \end{cases}$.

    Then $\begin{cases} (Df)^T\mathbb{1} = 0 \\ f^TDf = \mathrm{vol}(V) \\ f^TLf = \mathrm{vol}(V)\mathrm{NCut}(A,\bar{A}) \end{cases}$ enable us to rewrite the problem as

    $$\min_A f^TLf \text{ s.t. } \begin{cases} Df \perp \mathbb{1} \\ f^TDf = \mathrm{vol}(V) \end{cases} \text{ where } f_i = \begin{cases} \sqrt{\frac{\mathrm{vol}(\bar{A})}{\mathrm{vol}(A)}} & v_i \in A \\ -\sqrt{\frac{\mathrm{vol}(A)}{\mathrm{vol}(\bar{A})}} & v_i \in \bar{A} \end{cases}$$

    Then we relax the problem by allowing f to take arbitrary real values and the problem becomes

    $$\min_{f\in\mathbb{R}^n} f^TLf \text{ s.t. } \begin{cases} Df \perp \mathbb{1} \\ f^TDf = \mathrm{vol}(V) \end{cases}$$

    $\rightarrow$ substitute f with $g \equiv D^{1/2}f$. Then the problem becomes $\min_{g\in\mathbb{R}^n} g^TD^{-1/2}LD^{-1/2}g$ s.t. $\begin{cases} g \perp D^{1/2}\mathbb{1} \\ ||g||^2 = \mathrm{vol}(V) \end{cases}$, where $\begin{cases} D^{-1/2}LD^{-1/2} = L_{sym} \\ D^{1/2}\mathbb{1} \text{ is the first eigenvector of } L_{sym} \\ \mathrm{vol}(V) \text{ is constant} \end{cases}$. Thus by the Rayleigh–Ritz theorem, its solution g is given by the second eigenvector of L_{sym}. Resubstituting $f = D^{-1/2}g$ and using Proposition A.4.1, f is the second eigenvector of L_{rw} or equivalently the generalised eigenvector of the generalised eigenproblem $Lu = \lambda Du$.

  - $k > 2$

    Define the indicator vectors $h_j = (h_{1j},\dots,h_{nj})^T$ where $h_{ij} = \begin{cases} \frac{1}{\sqrt{\mathrm{vol}(A_j)}} & v_i \in A_j \\ 0 & \text{otherwise} \end{cases}$.

    Then set the matrix H as the matrix containing those k indicator vec-

tors as columns. Here $\begin{cases} H^T DH = I \\ h_i^T D h_i = 1 \\ h_i^T L h_i = \text{cut}(A_i, \bar{A}_i)/\text{vol}(A_i) \end{cases}$. By those relations, we can rewrite the problem as

$$\min_{A_1,\ldots,A_k} \text{Tr}(H^T LH) \text{ s.t. } H^T DH = I \text{ where } h_{ij} = \begin{cases} \frac{1}{\sqrt{\text{vol}(A_j)}} & v_i \in A_j \\ 0 & \text{otherwise} \end{cases}$$

Relaxing the discrete condition and substituting $T = D^{1/2}H$, we get the problem

$$\min_{T \in \mathbb{R}^{n \times k}} \text{Tr}(T^T D^{-1/2} L D^{-1/2} T) \text{ s.t. } T^T T = I$$

This is the standard trace minimisation problem which is solved by the matrix T which contains the first k eigenvectors of L_{sym} as columns. Re-substituting $H = D^{-1/2}T$ and by Proposition A.4.1, we see that the solution H consists of the first k eigenvectors of the matrix L_{rw} or the first k generalised eigenvectors of $Lu = \lambda Du$.

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